

```

timescale 1ns / 1ps
`default_nettype none
`include "parameters.vh"

```

```

module branch_decision(
    input wire [31:0] i_read_data1, // First operand
    input wire [31:0] i_read_data2, // Second operand
    input wire [2:0] i_Func3, // Determines branch type
    input wire [4:0] i_Funct5, // Determines atomic operation type
    input wire [6:0] i_opcode, // Opcode to distinguish branch vs AMO
    output reg o_Branch, // Branch decision
    output reg [31:0] o_AMO_result // Atomic operation result
);

```

```

always @(*) begin
    // Default outputs
    o_Branch = 1'b0;
    o_AMO_result = i_read_data1;

    if (i_opcode == `B) begin
        case(i_Func3)
            `BEQ: o_Branch = (i_read_data1 == i_read_data2);
            `BNE: o_Branch = (i_read_data1 != i_read_data2);
            `BLT: o_Branch = ($signed(i_read_data1) < $signed(i_read_data2));
            `BGE: o_Branch = ($signed(i_read_data1) >= $signed(i_read_data2));
            `BLTU: o_Branch = (i_read_data1 < i_read_data2);
            `BGEU: o_Branch = (i_read_data1 >= i_read_data2);
            default: o_Branch = 1'b0;
        endcase
    end
    else if (i_opcode == `AMO) begin
        case(i_Funct5)
            `AMO_ADD: o_AMO_result = i_read_data1 + i_read_data2;
            `AMO_SWAP: o_AMO_result = i_read_data2;
            `AMO_XOR: o_AMO_result = i_read_data1 ^ i_read_data2;
            `AMO_AND: o_AMO_result = i_read_data1 & i_read_data2;
            `AMO_OR: o_AMO_result = i_read_data1 | i_read_data2;
            `AMO_MIN: o_AMO_result = ($signed(i_read_data1) <
$signed(i_read_data2)) ? i_read_data1 : i_read_data2;
            `AMO_MAX: o_AMO_result = ($signed(i_read_data1) >
$signed(i_read_data2)) ? i_read_data1 : i_read_data2;
            `AMO_MINU: o_AMO_result = (i_read_data1 < i_read_data2) ? i_read_data1 :
i_read_data2;
            `AMO_MAXU: o_AMO_result = (i_read_data1 > i_read_data2) ? i_read_data1 :
i_read_data2;
            default: o_AMO_result = i_read_data1;
        endcase
    end
end

```

```
        endcase
    end
end

endmodule
```